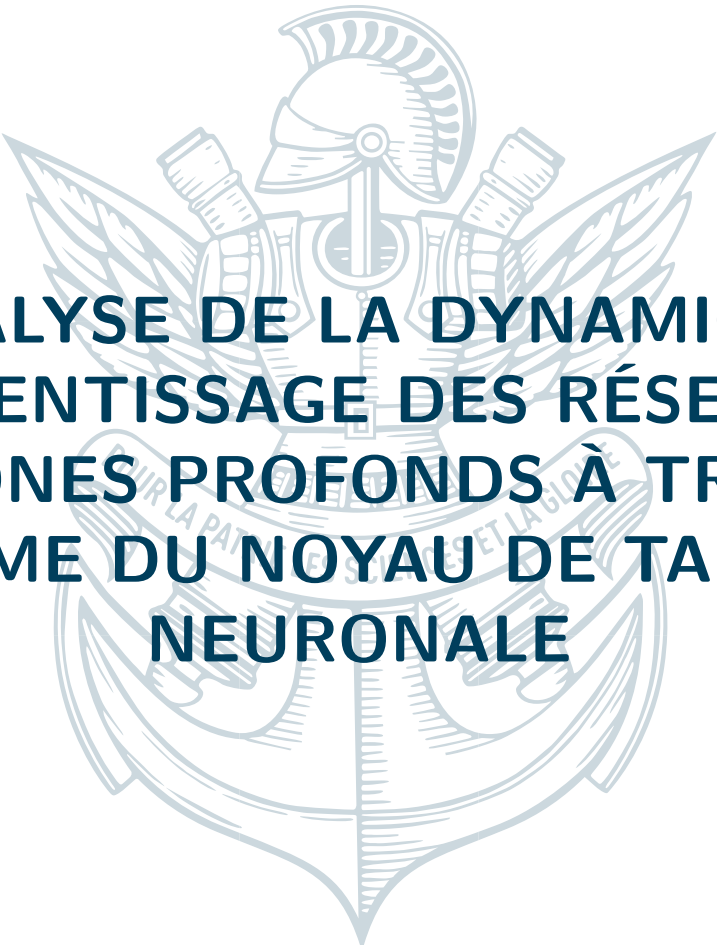




ANALYSE DE LA DYNAMIQUE D'APPRENTISSAGE DES RÉSEAUX DE NEURONES PROFONDS À TRAVERS LE PRISME DU NOYAU DE TANGENTE NEURONALE

**Contributions à la compréhension des corrections à largeur
finie et des régimes d'apprentissage**

20 août 2025

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RÉSUMÉ

Ce rapport présente les résultats d'un stage de recherche axé sur l'étude de la dynamique d'apprentissage des réseaux de neurones profonds. Nous explorons les mécanismes sous-jacents à travers le formalisme du Noyau de Tangente Neuronale (NTK), en portant une attention particulière aux corrections à large finie. Nos travaux s'articulent autour de la décomposition du NTK via l'entraînement de Sobolev, l'analyse de son espace à noyau reproduisant (RKHS) et l'étude des valeurs propres extrêmes. Nous présentons des bornes théoriques pour la plus petite valeur propre du NTK en nous appuyant sur des techniques de matrices aléatoires (RMT). Ces résultats sont ensuite appliqués à l'analyse de différentes architectures, notamment les réseaux de neurones à propagation avant (FCNN), les réseaux à mémoire matricielle (MMNN) et les transformeurs. Nous discutons également des implications de nos découvertes pour les applications en calcul scientifique (SciML) et proposons des pistes pour de futures recherches.

EXECUTIVE SUMMARY

(Executive summary placeholder)

INTRODUCTION

L'étude de la dynamique d'apprentissage dans les réseaux de neurones profonds constitue un domaine de recherche fondamental. Le régime du Noyau de Tangente Neuronale (NTK) a fourni un cadre puissant pour analyser le comportement des réseaux infiniment larges, où la dynamique d'entraînement se linéarise. Cependant, les réseaux pratiques ont une largeur finie, ce qui introduit des corrections complexes et un comportement d'apprentissage des caractéristiques (feature learning). Ce rapport vise à approfondir notre compréhension de ces phénomènes.

Nous commençons par une analyse théorique du NTK, en présentant sa décomposition basée sur l'entraînement de Sobolev et en caractérisant son RKHS. Une partie centrale de nos travaux est consacrée à l'étude du spectre du NTK, où nous cherchons à borner ses plus petites valeurs propres à l'aide d'outils issus de la théorie des matrices aléatoires. Ces résultats théoriques sont ensuite mis en perspective à travers l'analyse de plusieurs architectures : FCNN, MMNN et transformeurs. Pour les FCNN, nous étudions les corrections à largeur finie et le scaling des paramètres. Pour les MMNN et les transformeurs, nous discutons des défis spécifiques liés à l'optimisation et des intuitions sur leurs performances.

Enfin, nous explorons les applications de nos travaux, notamment en SciML, et discutons de l'alignement entre les hypothèses théoriques et les observations empiriques. Nous abordons également des perspectives inspirées de la physique statistique pour interpréter la convergence des modèles vers l'équilibre.

NOTATIONS

ReLU : La fonction d'activation ReLU (Rectified Linear Unit), définie par $\text{ReLU}(x) = \max(0, x)$.

$\mathbb{E}[\cdot]$: Espérance mathématique.

$\text{Var}(\cdot)$: Variance d'une variable aléatoire.

$\mathbb{I}_{\{\cdot\}}$: Fonction indicatrice, valant 1 si la condition est vraie, 0 sinon.

dx : Élément différentiel utilisé dans les intégrales.

1

CADRE THÉORIQUE DU NOYAU DE TANGENTE NEURONALE

1.1 ENTRAÎNEMENT DE SOBOLEV ET DÉCOMPOSITION DU NTK

(Présentation des théorèmes sur la décomposition du NTK via le Sobolev training. Description du RKHS pour le noyau de Sobolev et le noyau de Laplace.)

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INTRODUCTION : EIGENVALUE SCALING LAWS AND LEARNING DYNAMICS

The Neural Tangent Kernel (NTK) has emerged as a fundamental tool for understanding the training dynamics of deep neural networks. For a neural network $f(\cdot; \theta)$ with parameters θ , the NTK is defined as :

$$K^\infty(x_i, x_j) = \left\langle \frac{\partial f(i; \theta)}{\partial \theta}, \frac{\partial f(j; \theta)}{\partial \theta} \right\rangle$$

2.1 KEY OBJECTIVES

This document addresses three fundamental objectives :

1. **Eigenvalue scaling laws** : Derive decay rates $\mu_\ell \sim \ell^{-\alpha}$ for NTK operator eigenvalues
2. **Spectral impact on learning** : Understand how spectral properties determine learning dynamics
3. **Matrix vs. Operator relationship** : Analyze scaling laws for discrete matrix eigenvalues with respect to network depth l and data size n

2.2 NTK MATRIX VS. NTK OPERATOR

A crucial distinction exists between :

- **NTK Matrix** : Discrete sampled version $K \in \mathbb{R}^{n \times n}$ with entries $K_{ij} = K^\infty(x_i, x_j)$
- **NTK Operator** : Continuous integral operator $(\mathcal{L}f)(x) = \int K^\infty(x, y)f(y)dy$

Warning : The eigenvalues of the sampled NTK matrix are *not* the same as the NTK operator eigenvalues. The matrix spectrum provides discrete approximations that depend on the sampling strategy and data distribution.

2.3 INITIALIZATION : EDGE OF CHAOS

All analysis assumes Edge of Chaos (EOC) initialization :

- Initialize weights as $w_{ij} \sim \mathcal{N}(0, \sigma_w^2/\text{fan-in})$
- For ReLU networks : $\sigma_w^2 = 2$ to maintain unit variance through layers
- This ensures activations neither explode nor vanish with depth

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NTK MATRIX STRUCTURE AND SPECTRUM

3.1 EXAMPLE NTK MATRIX STRUCTURE

Consider a dataset of 3 points $x_1, x_2, x_3 \in \mathbb{R}^d$. The NTK matrix has entries :

$$K^\infty = \begin{pmatrix} k(x_1, x_1) & k(x_1, x_2) & k(x_1, x_3) \\ k(x_2, x_1) & k(x_2, x_2) & k(x_2, x_3) \\ k(x_3, x_1) & k(x_3, x_2) & k(x_3, x_3) \end{pmatrix}$$

For general depth l networks at EOC, the NTK kernel function is :

$$K^\infty(\mathbf{x}_1, \mathbf{x}_2) = \|\mathbf{x}_1\| \|\mathbf{x}_2\| \left(\sum_{k=1}^l \varrho^{\circ(k-1)}(\rho_1^\infty(\mathbf{x}_1, \mathbf{x}_2)) \right) \quad (1)$$

$$\times \prod_{k'=k}^{l-1} \varrho' \left(\varrho^{\circ(k'-1)}(\rho_1^\infty(\mathbf{x}_1, \mathbf{x}_2)) \right) \mathbf{I}_{m_l} \quad (2)$$

For 2-layer ReLU networks, this simplifies to :

$$k(x_i, x_j) = x_i^T x_j \cdot \arccos(-\langle x_i, x_j \rangle) + \sqrt{1 - \langle x_i, x_j \rangle^2}$$

3.2 NTK OPERATOR STRUCTURE

For functions $f, g \in L^2(\mathbb{S}^{d-1})$, the NTK operator acts as :

$$(K^\infty f)(x) = \int_{\mathbb{S}^{d-1}} k(x, y) f(y) d\sigma(y)$$

Key Properties :

- Symmetric positive definite operator
- Eigenfunctions are spherical harmonics $Y_{\ell, p}$
- Eigenvalues decay polynomially : $\mu_\ell \sim \ell^{-d}$

3.3 MATRIX SPECTRUM RESULTS

The key scaling laws for the discrete NTK matrix are :

Condition Number :

$$\kappa(K^\infty) \sim 1 + \frac{n}{3} + \mathcal{O}(n\xi/l)$$

The condition number grows linearly with data size n but improves with depth l .

Eigenvalue Distribution :

$$\lambda_{\min} \sim \frac{3l}{4}, \quad \lambda_{\max} \sim \frac{3nl}{4} \pm \xi \text{ where } \xi \sim \log(l)$$

The minimum eigenvalue scales linearly with depth, while the maximum scales with both depth and data size.

Key Insight : The training dynamics are dictated by the NTK matrix eigenvalues, not the operator eigenvalues. Deeper networks ($l \uparrow$) improve conditioning but with diminishing returns.

4

SOBOLEV TRAINING AND NTK SPECTRAL MODIFICATION

4.1 NTK-SOBOLEV OPERATOR FRAMEWORK

The key innovation in Sobolev training is the modification of the standard L^2 loss to incorporate high-order derivatives. The NTK-Sobolev operator P_s acts on the sphere $\mathbb{S}^{d-1}$ with the commutation property :

Spherical Harmonics Transform : Let $\mathcal{F}$ denote the mapping from $L^2(\mathbb{S}^d)$ to the spectral domain $\bigoplus_{\ell=0}^{\infty} \mathbb{C}^{N(d,\ell)}$, where $N(d,\ell)$ is the dimension of the ℓ -th spherical harmonic space.

The Sobolev operator P_s is defined through its action in Fourier space :

$$P_s = \sum_{\ell=0}^{\ell_{\max}} \sum_{p=1}^{N(d,\ell)} (1+\ell)^{2s} P_{\ell,p}$$

where $P_{\ell,p} = a_{\ell,p} a_{\ell,p}^T$ with spectral coefficients $(a_{\ell,p})_i = c_i Y_{\ell,p}(x_i)$.

4.2 FROM SOBOLEV LOSS TO DISCRETE MATRIX OPERATOR

Sobolev loss with gradients :

$$\mathcal{L}_s[f] = \|f\|_{L^2}^2 + \|\nabla f\|_{L^2}^2 = \int_{\mathbb{S}^d} f(x)^2 d\sigma(x) + \int_{\mathbb{S}^d} \|\nabla f(x)\|^2 d\sigma(x)$$

Fourier decomposition : For $f(x) = \sum_{\ell,p} \hat{f}_{\ell,p} Y_{\ell,p}(x)$:

$$\mathcal{L}_s[f] = \sum_{\ell=0}^{\infty} \sum_{p=1}^{N(d,\ell)} (1+\ell)^{2s} |\hat{f}_{\ell,p}|^2$$

Discretization : At points $\{x_i\}_{i=1}^n$:

$$\hat{f}_{\ell,p} \approx \sum_{i=1}^n c_i f(x_i) Y_{\ell,p}(x_i)$$

Final matrix form : $\mathcal{L}_s[f] \approx f^T P_s f$ where $f = (f(x_1), \dots, f(x_n))^T$.

4.3 SPECTRAL PROPERTIES AND DIMENSION GROWTH

The dimension of spherical harmonic spaces grows as :

$$N(d, \ell) \sim \frac{2\ell^{d-2}}{(d-2)!} \quad \text{as } \ell \rightarrow \infty$$

This exponential growth in dimension determines the computational complexity of the Sobolev operator discretization.

4.4 COMMON EIGENFUNCTIONS PROPERTY

[Spherical Harmonics as Common Eigenfunctions] For rotationally invariant kernels on $\mathbb{S}^{d-1}$:

- NTK operator K^∞ and Sobolev operator P_s share spherical harmonics $Y_{\ell,p}$ as eigenfunctions
- This is due to rotational invariance and Schur's lemma for irreducible representations
- Enables direct spectral modification analysis

[Commutation Property] The NTK operator K^∞ and Sobolev operator P_s commute :

$$[K^\infty, P_s] = 0$$

This holds for any sampling distribution $\rho(x)$ on $\mathbb{S}^{d-1}$ and implies that spectral decompositions are compatible.

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DEEP NTK ANALYSIS AND THEORETICAL RESULTS

5.1 NTK AT EDGE OF CHAOS

For MLPs with (a, b) -ReLU activation $\phi(s) = as + b|s|$, the Edge of Chaos initialization requires :

- Initialization : $\sigma^2 = (a^2 + b^2)^{-1}$
- Key parameter : $\Delta_\phi = \frac{b^2}{a^2 + b^2}$ (for standard ReLU : $\Delta_\phi = 0.5$)
- Cosine map : $\varrho(\rho) = \rho + \Delta_\phi \frac{2}{\pi} \left(\sqrt{1 - \rho^2} - \rho \arccos(\rho) \right)$

The Edge of Chaos is the unique initialization that remains invariant to network depth, preventing activation explosion or vanishing.

6

RECONCILING NTK MATRIX SPECTRUM AND SOBOLEV TRAINING

6.1 A UNIFIED SPECTRAL VIEWPOINT : COMMUTATION AND EIGENVALUE PRODUCTS

The reconciliation between the geometric view of the NTK (approximated by the inverse cosine distance matrix) and the functional view of Sobolev training lies in their shared algebraic structure.

- **Shared Invariance** : Both the NTK matrix K and the discrete Sobolev operator matrix P_s are rotationally invariant. This means their entries $(K)_{ij}$ and $(P_s)_{ij}$ depend only on the inner product $\langle x_i, x_j \rangle$.
- **Matrix Commutation** : As a consequence, both matrices can be expressed as functions of the Gram matrix G (where $G_{ij} = \langle x_i, x_j \rangle$). For two such matrices, $K = f(G)$ and $P_s = g(G)$, they commute :

$$KP_s = f(G)g(G) = g(G)f(G) = P_sK$$

This is the discrete analogue of the commutation of their underlying continuous operators.

- **Simultaneous Diagonalization** : Since they commute, K and P_s are simultaneously diagonalizable. They share the same set of eigenvectors, which are the eigenvectors of the Gram matrix G .
- **Product of Eigenvalues** : The spectrum of the Sobolev-modified NTK operator, KP_s , which governs the learning dynamics, is directly given by the product of the individual spectra :

$$\lambda_i(KP_s) = \lambda_i(K) \cdot \lambda_i(P_s)$$

(after aligning the eigenvector bases).

Key Insight : The challenge of analyzing the complex operator KP_s is reduced to separately analyzing the spectra of K and P_s . We can leverage the geometric approximation for K and combine it with a spectral analysis of P_s .

6.2 A PATH FORWARD : GEOMETRIC APPROXIMATION AND EXPERIMENTAL VALIDATION

Based on the unified spectral viewpoint, a clear research path emerges to develop a predictive model for Sobolev training dynamics.

Goal : Develop a semi-analytic model for the spectrum of the Sobolev-modified NTK matrix KP_s .

Proposed Strategy :

1. **Approximate the NTK Spectrum :** Use the established near-affine relationship between the NTK and the inverse cosine distance matrix W_l :

$$K \approx AW_l + B$$

The spectrum of the geometric matrix W_l can be analyzed to provide an approximation for the eigenvalues $\lambda_i(K)$.

2. **Analyze the Sobolev Operator Spectrum :** The matrix P_s is itself a kernel matrix with a well-defined kernel $p_s(t) = \sum_{\ell} (1 + \ell)^{2s} N(d, \ell) P_{\ell}(t)$, where P_{ℓ} are Legendre polynomials. Its spectrum $\lambda_i(P_s)$ can be characterized :
 - Analytically, through its kernel polynomial expansion.
 - Or by viewing it as a differential operator on the data manifold (graph), relating its spectrum to that of the graph Laplacian.
3. **Combine Spectra and Validate :** The core prediction is that the eigenvalues of the learning operator are given by the product of the component eigenvalues :

$$\lambda_i(KP_s) \approx \lambda_i(AW_l + B) \cdot \lambda_i(P_s)$$

Experimental Validation Plan :

- For a dataset on $\mathbb{S}^{d-1}$, numerically compute the matrices K , P_s , and the product KP_s .
- Compare the analytically predicted spectrum with the true, numerically computed spectrum of KP_s .
- Investigate the quality of this approximation as a function of the Sobolev parameter s , dimension d , data size n , and network depth l .

[The Sobolev Operator as a Kernel Matrix] The statement that the discrete Sobolev operator matrix P_s is itself a kernel matrix is a subtle but fundamental point. Here is a detailed derivation.

1. **Definition of the Matrix P_s** We start from its construction via projectors on the spherical harmonics :

$$(P_s)_{ij} = \left(\sum_{\ell=0}^{\ell_{\max}} \sum_{p=1}^{N(d, \ell)} (1 + \ell)^{2s} a_{\ell, p} a_{\ell, p}^{\top} \right)_{ij}$$

With the coefficient $(a_{\ell, p})_i = c_i Y_{\ell, p}(x_i)$, this becomes :

$$\begin{aligned} (P_s)_{ij} &= \sum_{\ell=0}^{\ell_{\max}} \sum_{p=1}^{N(d, \ell)} (1 + \ell)^{2s} (a_{\ell, p})_i (a_{\ell, p})_j \\ &= \sum_{\ell=0}^{\ell_{\max}} \sum_{p=1}^{N(d, \ell)} (1 + \ell)^{2s} (c_i Y_{\ell, p}(x_i)) (c_j Y_{\ell, p}(x_j)) \end{aligned}$$

2. Reorganization and Application of the Addition Theorem We can factor out the quadrature weights c_i, c_j and apply the spherical harmonic addition theorem, which is the key step.

$$(P_s)_{ij} = c_i c_j \sum_{\ell=0}^{\ell_{\max}} (1 + \ell)^{2s} \left[\sum_{p=1}^{N(d, \ell)} Y_{\ell, p}(x_i) Y_{\ell, p}(x_j) \right]$$

The addition theorem states that the term in brackets is a function only of the inner product $\langle x_i, x_j \rangle$:

$$\sum_{p=1}^{N(d, \ell)} Y_{\ell, p}(x) Y_{\ell, p}(y) = K_{\ell}(\langle x, y \rangle)$$

where $K_{\ell}(t) = \frac{N(d, \ell)}{\text{Area}(\mathbb{S}^{d-1})} P_{\ell}^{((d-2)/2)}(t)$ and $P_{\ell}^{(\lambda)}$ is a Gegenbauer polynomial.

3. Definition of the Kernel $p_s(t)$ By substituting this back, we can define a kernel function $p_s(t)$ that depends only on the inner product $t = \langle x_i, x_j \rangle$:

$$p_s(t) := \sum_{\ell=0}^{\ell_{\max}} (1 + \ell)^{2s} K_{\ell}(t)$$

Thus, the matrix element is indeed a weighted kernel evaluation :

$$(P_s)_{ij} = c_i c_j \cdot p_s(\langle x_i, x_j \rangle)$$

Conclusion The matrix P_s is a **weighted kernel matrix**. If the sampling is uniform, the weights c_i are constant and P_s becomes a standard kernel matrix (up to a scaling factor). This rotational invariance, shared with the NTK, is what guarantees that they commute.

6.3 DEEP NTK PROPERTIES

The limiting NTK at EOC for L -layer networks is given by :

$$K^{\infty}(\mathbf{x}_1, \mathbf{x}_2) = \|\mathbf{x}_1\| \|\mathbf{x}_2\| \left(\sum_{k=1}^l \varrho^{\circ(k-1)}(\rho_1^{\infty}(\mathbf{x}_1, \mathbf{x}_2)) \right) \quad (3)$$

$$\times \prod_{k'=k}^{l-1} \varrho' \left(\varrho^{\circ(k'-1)}(\rho_1^{\infty}(\mathbf{x}_1, \mathbf{x}_2)) \right) \Big) \mathbf{I}_{m_l} \quad (4)$$

Eigenvalue Decay : For L -layer ReLU networks with $L \geq 3$:

$$\mu_k \sim C(d, L) k^{-d}$$

where $C(d, L)$ depends on the parity of k and grows quadratically with L .

For the normalized NTK κ_{NTK}^L/L , the constant $C(d, L)$ grows linearly with L .

6.4 INVERSE COSINE DISTANCE MATRIX ANALYSIS

Inverse cosine distance matrix W_k for depth k :

$$W_{k i, i} = 0, \quad W_{k i_1, i_2} = \left(\frac{1 - \rho_k(x_{i_1}, x_{i_2})}{2} \right)^{-\frac{1}{2}} \text{ for } i_1 \neq i_2$$

Near-affine behavior :

- NTK matrix $K^\infty \approx A \cdot W_l + B$ (affine dependence)
- Spectral bounds transfer from W_k to NTK via this affine relationship
- Error terms : $O(k^{-1})$ - decreases with depth

This relationship enables indirect analysis of NTK spectral properties through simpler geometric matrices.

7

DEEP NARROW NEURAL NETWORKS

7.1 SCALED NTK AT INITIALIZATION

[Theorem 1 : Scaled NTK at Initialization] For f_θ^L initialized appropriately, as $L \rightarrow \infty$:

$$\tilde{\Theta}_0^L(x, x') \xrightarrow{p} \tilde{\Theta}^\infty(x, x')$$

where

$$\tilde{\Theta}^\infty(x, x') = (x^T x' + 1 + \mathbb{E}_g[\sigma(g(x))\sigma(g(x'))]) I_{d_{out}}$$

with $g \sim \text{GP}(0, \rho^2 d_{in}^{-1} x^T x' + \beta^2)$ (Gaussian random field).

Alternative formulation : $\kappa_1(\cos(u) \cdot v)$ where :

- $v = \frac{1}{1 + \beta^2 / \alpha^2}$, where β is the bias variance
- $\alpha = \frac{\|x\| \|x'\| \rho}{d_{in}}$, where $\cos(u)$ is the cosine distance between x, x'

This framework provides a promising direction for analyzing deep narrow networks through limited expansion analysis.

7.2 RESEARCH PERSPECTIVES FOR DEEP NARROW NETWORKS

Architectural modifications :

- **Unit concatenation** : Combine multiple narrow networks to increase expressivity
- **Skip connections** : Investigate ResNet-style connections :

$$\tilde{\Theta}_{\text{skip}}^\infty(x, x') = \tilde{\Theta}^\infty(x, x') + \text{skip terms}$$

Initialization studies :

- **Alternative initializations** : Explore different schemes beyond current setup

- $\beta \rightarrow 0$ **limit** : Analyze behavior when bias variance vanishes :

$$v = \frac{1}{1 + \beta^2/\alpha^2} \rightarrow 1 \text{ as } \beta \rightarrow 0$$

- **Complex kernel structures** : Find initializations that yield more intricate kernel forms

Theoretical extensions :

- Extension of Hayou & Yang's work on ResNets showing "wide and deep limits commute"
- Mean field analysis for deep narrow frameworks
- Careful analysis of stochasticity in initialization (not dropout)
- Experimental validation on practical tasks

7.3 ANALYSE SPECTRALE DU NTK ET BORNES SUR LES VALEURS PROPRES

(Discussion sur la plus grande valeur propre du NTK. Présentation des théorèmes de Terjek et du plan de preuve pour obtenir une borne sur la plus petite valeur propre en utilisant la RMT.)

8

MATHEMATICAL FOUNDATION : CONCENTRATION INEQUALITIES

8.1 SCALAR CONCENTRATION INEQUALITIES

The foundation of modern NTK analysis rests on powerful concentration inequalities that control the deviation of random variables from their expectations.

[Chernoff Bound for Bounded Random Variables] Let $X_1, \dots, X_n$ be independent random variables with $X_i \in [0, 1]$ and $\mathbb{E}[X_i] = \mu_i$. Let $S = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[S] = \sum_{i=1}^n \mu_i$. Then :

$$\Pr(S \geq (1 + \delta)\mu) \leq e^{-\frac{\delta^2 \mu}{2 + \delta}}$$

$$\Pr(S \leq (1 - \delta)\mu) \leq e^{-\frac{\delta^2 \mu}{2}}$$

for $\delta > 0$.

3[Bernstein's Inequality] Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}[X_i] = 0$, $|X_i| \leq M$, and $\mathbb{E}[X_i^2] \leq \sigma_i^2$. Let $S = \sum_{i=1}^n X_i$ and $\sigma^2 = \sum_{i=1}^n \sigma_i^2$. Then :

$$\Pr(|S| \geq t) \leq 2 \exp\left(-\frac{t^2/2}{\sigma^2 + Mt/3}\right)$$

8.2 QUADRATIC FORMS AND HANSON-WRIGHT INEQUALITIES

For neural networks, we frequently encounter quadratic forms in random variables, necessitating more sophisticated concentration tools.

[Classical Hanson-Wright Inequality] Let $Y_1, \dots, Y_n$ be independent mean-zero α -subgaussian random variables, and $\mathbf{A} = (a_{i,j})$ be a symmetric matrix. Then :

$$\Pr \left(\left| \sum_{i,j=1}^n a_{i,j} (Y_i Y_j - \mathbb{E}[Y_i Y_j]) \right| \geq t \right) \leq 2 \exp \left(-\frac{1}{C} \min \left\{ \frac{t^2}{\beta^4 \|\mathbf{A}\|_{HS}^2}, \frac{t}{\beta^2 \|\mathbf{A}\|_{op}} \right\} \right)$$

where $\|\mathbf{A}\|_{HS} = \sqrt{\sum_{i,j} |a_{i,j}|^2}$ is the Hilbert-Schmidt norm and $\|\mathbf{A}\|_{op}$ is the operator norm.

[Generalized Hanson-Wright for Random Tensors (Chang, 2022)] For random tensor vector $\overline{\mathcal{X}} \in (n \times I_1 \times \dots \times I_M) \times (I_1 \times \dots \times I_M)$ and fixed tensor $\overline{\mathcal{A}}$, the polynomial function :

$$f_j(\overline{\mathcal{X}}) = \left(\sum_{i,k=1}^n \mathcal{A}_{i,k} \star_M (\mathcal{X}_i - \mathbb{E}[\mathcal{X}_i]) \star_M (\mathcal{X}_k - \mathbb{E}[\mathcal{X}_k]) \right)^j$$

satisfies concentration bounds involving Ky Fan k -norms of tensor sums, extending classical matrix concentration to tensor settings.

8.3 MATRIX CONCENTRATION INEQUALITIES

Matrix concentration inequalities provide direct control over eigenvalues of random matrix sums, which is essential for NTK analysis.

[Matrix Chernoff Bound (Tropp, 2012)] Let $X_1, \dots, X_n$ be independent random Hermitian matrices with $X_i \preceq R \cdot I$ and $\mathbb{E}[X_i] \preceq \mu \cdot I$. Let $S = \sum_{i=1}^n X_i$. Then :

$$\Pr(\lambda_{\max}(S) \geq (1 + \delta)\mu n) \leq d \left(\frac{e^\delta}{(1 + \delta)^{1+\delta}} \right)^{\mu n/R}$$

[Matrix Bernstein Inequality (Vershynin, 2018)] Let $X_1, \dots, X_n$ be independent random matrices with $\mathbb{E}[X_i] = 0$, $\|X_i\| \leq L$, and $\|\sum_{i=1}^n \mathbb{E}[X_i X_i^*]\| \leq \sigma^2$. Then :

$$\Pr \left(\left\| \sum_{i=1}^n X_i \right\| \geq t \right) \leq 2d \exp \left(-\frac{t^2/2}{\sigma^2 + Lt/3} \right)$$

9

FUNDAMENTAL MATHEMATICAL TOOLS

9.1 MATRIX ANALYSIS TECHNIQUES

[Weyl's Inequality] For Hermitian matrices A, B :

$$\lambda_i(A) + \lambda_n(B) \leq \lambda_i(A + B) \leq \lambda_i(A) + \lambda_1(B)$$

[Schur Product Theorem] For PSD matrices P, Q :

$$P \circ Q \geq P \min_i Q_{ii}$$

[Gershgorin Circle Theorem] For matrix $A = (a_{ij})$, all eigenvalues lie in the union of discs :

$$\bigcup_{i=1}^n \left\{ z \in \mathbb{C} : |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}| \right\}$$

9.2 SPECIALIZED NEURAL NETWORK TOOLS

[Khatri-Rao Product] For matrices $A \in \mathbb{R}^{m \times k}$ and $B \in \mathbb{R}^{n \times k}$, the Khatri-Rao product $A \star B \in \mathbb{R}^{mn \times k}$ is the column-wise Kronecker product.

[Hadamard Power] For matrix A and positive integer r , the r -th Hadamard power $A^{\odot r}$ is the element-wise r -th power.

10

PROOF SCHEMES AND HIGH-LEVEL STRATEGIES

10.1 GENERAL FRAMEWORK FOR EIGENVALUE BOUNDS

All modern approaches to bounding $()$ follow a common high-level structure, though they differ in technical details and specific tools employed.

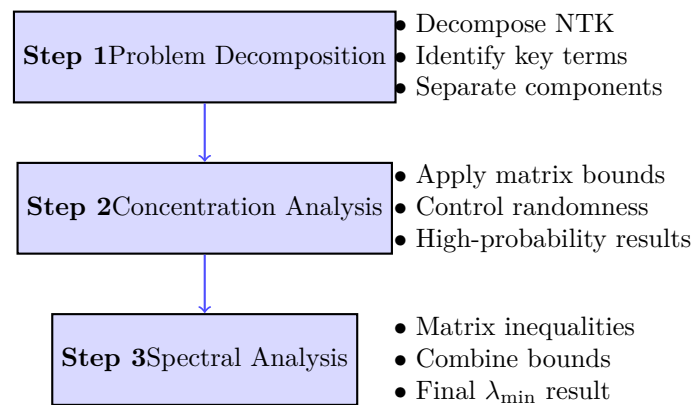


FIGURE 1 – General Framework for NTK Eigenvalue Bounds

10.2 PROOF SCHEME 1 : EMPIRICAL NTK DECOMPOSITION (NGUYEN ET AL.)

High-Level Strategy : Direct decomposition of empirical NTK $\bar{K}^{(L)} = JJ^T$ using feature matrices and chain rule to obtain precise bounds on .

10.2.1 • STEP 1 : FUNDAMENTAL CHAIN RULE DECOMPOSITION

The proof begins by decomposing the empirical NTK $\bar{K}^{(L)} = JJ^T$ where J is the Jacobian matrix (??). Applying the chain rule and standard algebraic manipulations yields the fundamental decomposition :

$$\bar{K}^{(L)} = JJ^T = \sum_{k=0}^{L-1} F_k F_k^T \circ B_{k+1} B_{k+1}^T \quad (5)$$

where :

- $F_k = [f_k(x_1), \dots, f_k(x_N)]^T \in N \times n_k$ are the **feature matrices** at layer k
- $B_k \in N \times n_k$ are the **derivative term matrices** with i -th row given by :

$$(B_k)_i = \begin{cases} \Sigma_k(x_i) \left(\prod_{l=k+1}^{L-1} W_l \Sigma_l(x_i) \right) W_L, & k \in [L-2] \\ \Sigma_{L-1}(x_i) W_L, & k = L-1 \\ \frac{1}{\sqrt{N}} \mathbf{1}_N, & k = L \end{cases} \quad (6)$$

where $\Sigma_k(x) = \text{diag}([\sigma'(g_{k,j}(x))]_{j=1}^{n_k})$ is the diagonal matrix of activation derivatives.

10.2.2 • STEP 2 : APPLICATION OF SCHUR PRODUCT THEOREM AND WEYL'S INEQUALITY

For PSD matrices $P, Q \in n \times n$, the Schur product theorem :

$$P \circ Q \geq P \min_{i \in [n]} Q_{ii} \quad (7)$$

Applying this to each term, then using Weyl's inequality for the sum :

$$JJ^T \geq \sum_{k=0}^{L-1} F_k F_k^T \circ B_{k+1} B_{k+1}^T \quad (8)$$

$$\geq \sum_{k=0}^{L-1} F_k F_k^T \min_{i \in [N]} \|(B_{k+1})_i\|_2^2 \quad (9)$$

This reduction separates the problem into :

1. The **minimum eigenvalues of feature Gram matrices** : $F_k F_k^T$
2. The **minimum row norms of derivative matrices** : $\min_{i \in [N]} \|(B_{k+1})_i\|_2^2$

10.2.3 • STEP 3 : BOUNDING FEATURE MATRIX EIGENVALUES

To bound $F_k F_k^T$, the strategy uses Chernoff-type matrix concentration. The fundamental bound is given by : [Matrix Concentration for Features] Let $\lambda = \mathbb{E}_{w \sim \mathcal{N}(0, \beta_k^2 \mathbf{I}_{n_{k-1}})} [\sigma(F_{k-1} w) \sigma(F_{k-1} w)^T]$. If

$$n_k \geq \max \left(N, cQ \max(1, \log(4Q)) \log \frac{N}{\delta} \right) \quad (10)$$

where $Q = \frac{\beta_k^2 \|F_{k-1}\|_F^2}{\lambda}$, then with probability at least $1 - \delta$:

$$F_k^2 \geq \frac{n_k \lambda}{4} \quad (11)$$

This lemma assumes that :

- The width n_k of layer k is sufficiently large compared to the number of samples N
- The feature matrix F_{k-1} from the previous layer has bounded Frobenius norm

These conditions ensure concentration of the minimum singular value of the feature matrix around its expected value.

10.2.4 • STEP 4 : SPECTRAL ANALYSIS VIA HERMITE EXPANSION

To bound λ , we use the Hermite expansion of the ReLU. and exploiting the homogeneity of σ (unless, generalized hermite) :

$$\lambda = \mathbb{E}[\sigma(F_{k-1} w) \sigma(F_{k-1} w)^T] \quad (12)$$

$$= D \mathbb{E}[\sigma(\hat{F}_{k-1} w) \sigma(\hat{F}_{k-1} w)^T] D \quad (13)$$

where $D = \text{diag}(\|\{F_{k-1}\}_i\|_2)$ and $\hat{F}_{k-1} = D^{-1} F_{k-1}$.

Applying the Hermite expansion $\sigma(x) = \sum_{r=0}^{\infty} \mu_r(\sigma) H_r(x)$:

$$\mathbb{E}[\sigma(\hat{F}_{k-1} w) \sigma(\hat{F}_{k-1} w)^T] = \mu_0(\sigma)^2 \mathbf{1}_N \mathbf{1}_N^T + \sum_{s=1}^{\infty} \mu_s(\sigma)^2 (\hat{F}_{k-1}^{*s}) (\hat{F}_{k-1}^{*s})^T \quad (14)$$

where $\hat{F}_{k-1}^{*s}$ represents the s -th Khatri-Rao power of $\hat{F}_{k-1}$.

10.2.5 • STEP 5 : REDUCTION TO KHATRI-RAO POWERS

For an integer $r \geq 2$, we can lower bound :

$$\lambda \geq \mu_r(\sigma)^2 D (\hat{F}_{k-1}^{*r}) (\hat{F}_{k-1}^{*r})^T D = \mu_r(\sigma)^2 \frac{(F_{k-1}^{*r}) (F_{k-1}^{*r})^T}{\max_{i \in [N]} \|(F_{k-1})_i\|_2^{2(r-1)}} \quad (15)$$

10.2.6 • STEP 6 : FEATURE CENTERING AND HADAMARD POWER ANALYSIS

To analyze $(F_{k-1}^{*r}) (F_{k-1}^{*r})^T = (F_{k-1} F_{k-1}^T)^{or}$, we introduce centered features $\tilde{F}_{k-1} = F_{k-1} - \mathbb{E}_X[F_{k-1}]$.

Let $\mu = \mathbb{E}_x[f_{k-1}(x)] \in \mathbb{R}^{n_{k-1}}$ and $\Lambda = \text{diag}(F_{k-1} \mu - \|\mu\|_2^2 \mathbf{1}_N)$. Then :

$$(F_{k-1}^{*r}) (F_{k-1}^{*r})^T = (F_{k-1} F_{k-1}^T)^{or} \succeq \left(\tilde{F}_{k-1} \tilde{F}_{k-1}^T - \frac{\Lambda \mathbf{1}_N \mathbf{1}_N^T \Lambda}{\|\mu\|_2^2} \right)^{or} \quad (16)$$

10.2.7 • STEP 7 : APPLICATION OF GERSHGORIN CIRCLE THEOREM

To bound the minimum eigenvalue of $(\tilde{F}_{k-1}\tilde{F}_{k-1}^T)^{or}$, we apply the Gershgorin circle theorem :

$$(\tilde{F}_{k-1}^{*r})(\tilde{F}_{k-1}^{*r})^T \geq \min_{i \in [N]} \|(\tilde{F}_{k-1})_{i:}\|_2^{2r} - N \max_{i \neq j} |\langle (\tilde{F}_{k-1})_{i:}, (\tilde{F}_{k-1})_{j:} \rangle|^r \quad (17)$$

Using concentration properties of centered features, we show :

$$\|(\tilde{F}_{k-1})_{i:}\|_2^{2r} = \Theta \left(\left(d \prod_{l=1}^{k-1} n_l \beta_l^2 \right)^r \right) \quad (18)$$

$$N \max_{i \neq j} |\langle (\tilde{F}_{k-1})_{i:}, (\tilde{F}_{k-1})_{j:} \rangle|^r = o \left(\left(d \prod_{l=1}^{k-1} n_l \beta_l^2 \right)^r \right) \quad (19)$$

with exponentially high probability.

10.2.8 • STEP 8 : BOUNDING DERIVATIVE TERM NORMS, VERY TECHNICAL

To bound $\min_{i \in [N]} \|(B_{k+1})_{i:}\|_2^2$, we analyze each case :

For $k \in [L-2]$:

$$\|(B_{k+1})_{i:}\|_2^2 = \|\Sigma_{k+1}(x_i) \left(\prod_{l=k+2}^{L-1} W_l \Sigma_l(x_i) \right) W_L\|_2^2 \quad (20)$$

$$= \Theta \left(\beta_L^2 n_{k+1} \prod_{l=k+2}^{L-1} n_l \beta_l^2 \right) \quad (21)$$

This bound uses some very technical and important tools :

- **Operator norm for concentration of Gaussian matrices**
- **Inductive analysis of random matrix products for every layer**
- **Properties of derivative matrices $\Sigma_l(x)$ for every layer**

10.2.9 • STEP 9 : FINAL ASSEMBLY AND WIDTH OPTIMIZATION

Combining all bounds via union bound yields :

$$\bar{K}^{(L)} \geq \sum_{k=0}^{L-1} \mu_r(\sigma)^2 \Theta \left(d \prod_{l=1}^k n_l \beta_l^2 \right) \Theta \left(\beta_L^2 \prod_{l=k+1}^{L-1} n_l \beta_l^2 \right) \quad (22)$$

$$= \mu_r(\sigma)^2 \Theta \left(d \beta_L^2 \sum_{k=0}^{L-1} \prod_{l=1}^{L-1} n_l \beta_l^2 \right) \quad (23)$$

The width condition requires at least one layer to satisfy :

$$n_k = \tilde{\Omega}(N \log N), \quad \text{with } \xi_k = 1 \quad (24)$$

where ξ_k indicates layer k is "wide".

10.2.10 • UPPER BOUND VIA DIAGONAL ANALYSIS

For the upper bound, we use :

$$JJ^T \leq (JJ^T)_{11} = \sum_{k=0}^{L-1} \|(F_k)_{1:}\|_2^2 \|(B_{k+1})_{1:}\|_2^2 \quad (25)$$

$$= \sum_{k=0}^{L-1} \|f_k(x_1)\|_2^2 \|(B_{k+1})_{1:}\|_2^2 \quad (26)$$

$$= \mathcal{O} \left(d\beta_L^2 \sum_{k=0}^{L-1} \prod_{l=1}^{L-1} n_l \beta_l^2 \right) \quad (27)$$

10.2.11 • FINAL RESULT

Assuming $\beta_l = 1$ for simplicity and that some layer k^* satisfies $n_{k^*} = \tilde{\Omega}(N)$, the main theorem gives :

$$\mu_r(\sigma)^2 \Omega(d) \leq \bar{K}^{(L)} \leq \mathcal{O}(dL) \quad (28)$$

with probability at least $1 - \text{poly}(N) \exp(-\Omega(\min_l n_l))$.

This approach significantly reduces width requirements compared to classical methods, requiring only one wide layer rather than all layers.

11

CORRECTIONS À LARGEUR FINIE ET RÉGIMES D'APPRENTISSAGE

11.1 DÉVELOPPEMENT THÉORIQUE POUR LES FCNN

(Présentation des théorèmes et formules pour les corrections à largeur finie dans les FCNN. Analyse des lois de scaling du reste et discussion sur les termes d'erreur, en lien avec les ensembles de matrices aléatoires QUE GOE.)

11.2 APPROCHES ALTERNATIVES ET CONCENTRATION

(Explication de l'approche alternative pour les FCNN via les réseaux paraboliques de Terjek, permettant de se placer en régime de feature learning tout en conservant des propriétés du NTK. Discussion sur les bornes de concentration.)

12

ANALYSE D'ARCHITECTURES SPÉCIFIQUES

12.1 RÉSEAUX À MÉMOIRE MATRICIELLE (MMNN) ET TRANSFORMEURS

(Description des techniques de préconditionnement du NTK pour ces architectures. Discussion sur les régimes "feature" et "kernel", et sur les défis liés à l'optimisation. Présentation d'une conjecture sur la stabilité autour des minima globaux.)

13

MMNN NTK FOR 2 HIDDEN LAYERS

13.1 RECURSIVE

For a 2 hidden layer MMNN, we can write the recursive formulation of the NTK kernel. Let's recall the general recursive formula for layer ℓ :

suppose that you have a low rank hidden layer, with a width r

so the architecture is :

$$\begin{array}{ccccccc} (x, y) & \xrightarrow{\sigma W_1} & h_1 & \xrightarrow{A_1 \cdot + c_1} & r & \xrightarrow{\sigma W_2} & h_2 \xrightarrow{A_2 \cdot + c_2} (x_2, y_2) \\ & & N & & (x_1, y_1) \text{ in } \mathbb{R}^r & & N \end{array}$$

And here N grow to infinity, and r is fixed.

Then the NTK random kernel is :

$$\Theta^{(\ell)}(x, x') = \Theta^{(\ell-1)}(x, x') \cdot \left(\sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x') \right) + \Sigma^{(\ell)}(x, x') + 1 \quad (29)$$

For $\ell = 2$, starting from $\Theta^{(0)} = 0$, we have :

$$\begin{aligned} \Theta^{(1)} &= \mathbb{E}_{w,b} [\sigma(w^\top x + \beta b) \sigma(w^\top x' + \beta b)] + 1 \\ \Theta^{(2)} &= \Theta^{(1)} \cdot \left(\sigma_A^2 \mathbb{E}_{w,b} \left[\dot{\sigma}(w^\top h^{(1)}(x) + \beta b) \dot{\sigma}(w^\top h^{(1)}(x') + \beta b) w^\top w \right] \right) \\ &\quad + \mathbb{E}_{w,b} [\sigma(w^\top h^{(1)}(x) + \beta b) \sigma(w^\top h^{(1)}(x') + \beta b)] + 1 \end{aligned}$$

Here, $\Theta^{(1)}$ is a deterministic NNGP kernel (Gaussian Process) since it depends only on the input x , while $\Theta^{(2)}$ is a random field (Deep Gaussian Process) that depends on the realization of the first layer's output $h^{(1)}$. When

$\beta = 0$, these expectations simplify by removing the bias terms b . This non-Gaussian nature of the kernels for $\ell > 1$ is a key distinguishing feature of MMNNs compared to standard neural networks.

When $\beta = 0$, the equations simplify to :

Main focus : NTK kernels with β

$$\begin{aligned}\Theta^{(1)} &= \mathbb{E}_w [\sigma(w^\top x) \sigma(w^\top x')] + 1 \\ \Theta^{(2)} &= \Theta^{(1)} \cdot \left(\sigma_A^2 \mathbb{E}_w [\dot{\sigma}(w^\top h^{(1)}(x)) \dot{\sigma}(w^\top h^{(1)}(x')) w^\top w] \right) \\ &\quad + \mathbb{E}_w [\sigma(w^\top h^{(1)}(x)) \sigma(w^\top h^{(1)}(x'))] + 1\end{aligned}$$

Note that for stable signal propagation at the Edge of Chaos (EOC), we require for 1 rank hidden layer :

$$\sigma_A^2 \cdot \text{Tr}(\Sigma_w) = 2$$

Then for a rank r

$$\sigma_A^2 \cdot \text{Tr}(\Sigma_w) = \frac{2}{r}$$

And we assume at least isotropic gaussian for the weights without loss of generality and to ensure fast computations.

This condition ensures the signal neither vanishes nor explodes as it propagates through the network layers. This also means we need to normalize by $\sqrt{r}$ the output of the first MMNN layer. We call x_1 and y_1 the outputs of the first layer unnormalized.

We recall also that $\Sigma^{(1)}$ is the arccosine kernel, which has a closed form in function of the correlation ρ between x and x' :

$$\Sigma^{(1)}(\rho) = \frac{1}{\pi} \left(\sqrt{1 - \rho^2} + \rho(1 - \arccos(\rho)) \right)$$

We recall also that $\dot{\Theta}^{(1)}(\rho) = \mathbb{E}_w [w^\top w \dot{\sigma}(w^\top x) \dot{\sigma}(w^\top x')]$. We can compute this exactly when w is isotropic gaussian with variance 1 :

In fact a simple computation takes the expectation of the product of the two derivatives of the ReLU, by independence of $\|w\|^2$ and $\mathbb{1}_{w^\top x > 0} \cdot \mathbb{1}_{w^\top x' > 0}$:

$$\begin{aligned}\dot{\Theta}^{(1)}(\rho) &= \mathbb{E}_w [w^\top w \dot{\sigma}(w^\top x) \dot{\sigma}(w^\top x')] \\ &= \mathbb{E}_w [w^\top w \cdot \mathbb{1}_{w^\top x > 0} \cdot \mathbb{1}_{w^\top x' > 0}] \\ &= \mathbb{E}[\|w\|^2] \cdot \mathbb{P}(w^\top x > 0, w^\top x' > 0) \\ &= \text{Tr}(\Sigma_w) \left(\frac{1}{2} - \frac{\arccos(\rho)}{2\pi} \right)\end{aligned}$$

It is just a reformulation of the fact that we integrate over a cone of angle $\arccos(\rho)$ a radial function.

We then write $\Theta^{(2)}$ as a function of ρ and ρ_1 :

$$\begin{aligned}\Theta^{(2)}(\rho, \rho_1) &= \Theta^{(1)}(\rho) \cdot \sigma_A^2 \text{Tr}(\Sigma_w) \left(\frac{1}{2} - \frac{\arccos(\rho)}{2\pi} \right) + \frac{1}{r} \cdot \Sigma^{(1)}(\rho_1) \|x_1\| \|y_1\| + 1 \\ \text{with } \Sigma^{(1)}(\rho) &= \frac{1}{\pi} \left(\sqrt{1 - \rho^2} + \rho(1 - \arccos(\rho)) \right)\end{aligned}$$

where ρ_1 is the correlation between the outputs of the first layer, that is a random variable and ρ is the correlation between the inputs.

Note that this multiplicative value is between 0 and 1, that is we can compute a lot of stuff after concatenating layers and getting some exponential scaling laws. We emphasize the fact we multiply by $\|x_1\|\|y_1\|$ which is a random variable, but as we will see, it is independent of the correlation ρ_1 .

13.2 EXPRESSION FOR RANDOM VARIABLES $\rho_1, \|x_1\|^2, \|y_1\|^2$

The output of the 1st layer is a gaussian process of dimension r , where each coordinate is a gaussian process with covariance matrix :

$$\Sigma_i^{(1)} = \mathbb{E}_w [\sigma(w^\top x) \sigma(w^\top x')]$$

and the coordinates are ρ -correlated. If we stack horizontally the outputs x_1 and y_1 in a vector of dimension $2r$, the full $2r \times 2r$ covariance matrix is therefore :

$$\Sigma^{(1)} = \begin{pmatrix} I & \rho I \\ \rho I & I \end{pmatrix}$$

It's a kronecker product of r matrices I and $2d$ ρ correlated gaussian variables.

it happens, by a magic, that we can compute exactly the joint distribution of the random variable $(\rho_1, \|x_1\|^2, \|y_1\|^2)$:

$$\rho_1 = \frac{\sum_{i=1}^r x_i y_i}{\sqrt{\sum_{i=1}^r x_i^2} \sqrt{\sum_{i=1}^r y_i^2}}$$

The squared norms x_1^2 and y_1^2 follow a bivariate Kibble distribution and their cosine distance follows an independent Fisher like distribution. The Kibble distribution is defined for $u, v \geq 0$ and $\rho \in [-1, 1]$, while the Fisher distribution is defined for $\theta \in [-1, 1]$ and $\rho \in [-1, 1]$ with $r > 2$.

$$\begin{aligned} (\|x_1\|^2, \|y_1\|^2) &\sim \text{Kibble}(r, \rho) \\ &= \frac{(uv)^{\frac{r/2-1}{2}} e^{-\frac{u+v}{2(1-\rho^2)}}}{\Gamma(r/2)(2(1-\rho^2))^{r/2+1} \rho^{\frac{r/2-1}{2}}} \cdot I_{r/2-1} \left(\frac{\rho \sqrt{uv}}{1-\rho^2} \right) \\ \rho_1 &\sim \text{Fisher}(\rho, r) \\ &= \frac{(r-2)\Gamma(r-1)(1-\rho^2)^{\frac{r-1}{2}}(1-\theta^2)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(1-\rho\theta)^{r-\frac{3}{2}}} \times {}_2F_1 \left(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; \frac{1+\rho\theta}{2} \right) \end{aligned}$$

where $u = \|x_1\|^2$, $v = \|y_1\|^2$, $\theta = \cos(x_1, y_1)$, and I_α is the modified Bessel function of the first kind.

and the correlation of x, y :

where x_{1i} and y_{1i} are ρ -correlated gaussian variables.

We ensure that the reader has understood that the correlation of x, y is independent of the squared norms of x, y !

13.2.1 • FINAL RESULT

We can then state the final result : For 2 entries x, y in $\mathbb{R}^d$ and a low rank r hidden layer, the NTK for a 2 hidden layer MMNN is :

$$\Theta^{(2)}(x, y) = \Theta^{(1)}(\rho) \cdot \left(1 - \frac{\arccos(\rho_1)}{\pi}\right) \frac{1}{r} + \frac{1}{r} \Sigma^{(1)}(\rho_1) \|x_1\| \|y_1\| + 1$$

with $\Sigma^{(1)}(\rho) = \frac{1}{\pi} \left(\sqrt{1 - \rho^2} + \rho(1 - \arccos(\rho)) \right)$

where the random variables follow :

- $\rho_1 \sim \text{Fisher}(\rho, r)$ is independent of the norms
- $(\|x_1\|^2, \|y_1\|^2) \sim \text{Kibble}(r, \rho)$ are chi-squared variables with r degrees of freedom from correlated gaussian variables.

13.3 INTUITIONS SUR LES PERFORMANCES

(Discussion finale basée sur les résultats théoriques et expérimentaux pour expliquer les performances observées pour les MMNN, les transformeurs et les ResNets.)

14

RÉSULTATS EXPÉRIMENTAUX ET APPLICATIONS

14.1 VALIDATION DES LOIS DE SCALING ET DES CORRECTIONS

(Présentation des expériences numériques validant les lois de scaling pour les corrections à largeur finie dans les FCNN.)

14.2 ANALYSE HESSIENNE ET DYNAMIQUE

(Présentation des expériences sur l'approche hessienne du NTK et l'étude des termes de type Lyapunov.)

14.3 APPLICATIONS EN SCIML ET DISCUSSION

(Discussion sur l'application des résultats en Scientific Machine Learning. Vérification des hypothèses théoriques sur les grands modèles et le scaling des paramètres. Lien avec la physique statistique et la convergence vers l'équilibre thermique.)

ANNEXES



001504621_896x598_c.png

FIGURE 2 – Aperçu de l'interface mise à disposition en source ouverte pour l'extraction de connaissances

A

FISHER, KIBBLE DISTRIBUTIONS AND THE FAKE "CURSE OF DIMENSIONALITY"

A.1 DISTRIBUTION OF THE CORRELATION COEFFICIENT r

Fisher derived the exact law of r in 1915, representing the cosine between Gaussian vectors of dimension r with correlation ρ :

$$p(r|\rho, r) = \frac{(r-2)\Gamma(r-1)(1-\rho^2)^{\frac{r-1}{2}}(1-r^2)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(1-\rho r)^{r-\frac{3}{2}}} \times {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho r}{2}\right)$$

Where Γ is the gamma function and ${}_2F_1$ is the hypergeometric function.

For large r , Fisher's z-transform gives $z = \operatorname{arctanh}(r) \sim \mathcal{N}\left(\operatorname{arctanh}(\rho) + \frac{\rho}{2(r-1)}, \frac{1}{r-3}\right)$. The variance of r decreases as $O(1/r)$. When $\rho \sim O(1/\sqrt{r})$, geometric noise masks correlation.

A.2 THE KIBBLE DISTRIBUTION

The joint law of correlated chi-squared variables $U = \|X\|^2$ and $V = \|Y\|^2$ follows Kibble's 1941 distribution :

$$f(u, v) = \frac{(uv)^{\frac{r/2-1}{2}} e^{-\frac{u+v}{2(1-\rho^2)}}}{\Gamma(r/2)(2(1-\rho^2))^{r/2+1} \rho^{\frac{r/2-1}{2}}} \cdot I_{r/2-1}\left(\frac{\rho\sqrt{uv}}{1-\rho^2}\right)$$

Where Γ is the gamma function and I_α is the modified Bessel function of the first kind.

Mean $\mathbb{E}[U] = r$ and variance $\operatorname{Var}(U) = 2r$ grow linearly. U/r converges to $\mathcal{N}(1, 2/r)$ with polynomial concentration.

B

PRICE'S THEOREM AND DERIVATIONS

Price's theorem was a central tool in our analysis.

• STATEMENT OF PRICE'S THEOREM

Suppose that X_1 and X_2 forms a gaussian vector with joint distribution $f_{X_1, X_2}(x_1, x_2)$. Then, for any function $g(x_1, x_2)$, we have :

$$\mathbb{E}[g(X_1, X_2)] = \int_0^\rho \mathbb{E}\left[\frac{\partial^2 g}{\partial x_1 \partial x_2}(X_1, X_2)\right] d\rho + \text{Constant}$$

where the constant is the value of the expectation when $\rho = 0$.

• **CALCULATION OF I_{12} AND I_{21} (FIRST-ORDER MOMENTS)**

For I_{12} , we need to calculate :

$$I_{12} = \int_a^\infty \int_b^\infty z_2 \phi_2(z_1, z_2; \rho) dz_2 dz_1$$

Step 1 : Inner integral We first focus on the inner integral with respect to z_2 . The exact formula for the conditional density is :

$$\phi_2(x, y; \rho) = \frac{1}{\sqrt{1-\rho^2}} \phi_1(y) \phi_1\left(\frac{x-\rho y}{\sqrt{1-\rho^2}}\right)$$

Applying this formula with $x = z_1$ and $y = z_2$:

$$\phi_2(z_1, z_2; \rho) = \frac{1}{\sqrt{1-\rho^2}} \phi_1(z_2) \phi_1\left(\frac{z_1-\rho z_2}{\sqrt{1-\rho^2}}\right)$$

The inner integral becomes :

$$\int_b^\infty z_2 \frac{1}{\sqrt{1-\rho^2}} \phi_1(z_2) \phi_1\left(\frac{z_1-\rho z_2}{\sqrt{1-\rho^2}}\right) dz_2$$

Step 2 : Change of variable Let $u = \frac{z_1-\rho z_2}{\sqrt{1-\rho^2}}$. Then :

$$z_2 = \frac{z_1 - u\sqrt{1-\rho^2}}{\rho}$$

$$dz_2 = -\frac{\sqrt{1-\rho^2}}{\rho} du$$

The lower bound becomes $u_{max} = \frac{z_1-\rho b}{\sqrt{1-\rho^2}}$. The integral transforms into :

$$\frac{1}{\sqrt{1-\rho^2}} \int_{-\infty}^{\frac{z_1-\rho b}{\sqrt{1-\rho^2}}} \left(\frac{z_1 - u\sqrt{1-\rho^2}}{\rho}\right) \phi_1\left(\frac{z_1 - u\sqrt{1-\rho^2}}{\rho}\right) \phi_1(u) \left(-\frac{\sqrt{1-\rho^2}}{\rho}\right) du$$

Step 3 : Simplification Grouping terms and using properties of ϕ_1 :

$$-\frac{1}{\rho^2} \int_{-\infty}^{\frac{z_1-\rho b}{\sqrt{1-\rho^2}}} (z_1 - u\sqrt{1-\rho^2}) \phi_1(u) \phi_1\left(\frac{z_1 - u\sqrt{1-\rho^2}}{\rho}\right) du$$

Step 4 : Using the conditional density formula The exact formula is :

$$\phi_2(x, y; \rho) = \frac{1}{\sqrt{1-\rho^2}} \phi_1(x) \phi_1\left(\frac{y-\rho x}{\sqrt{1-\rho^2}}\right)$$

Applying it correctly to our case with $x = z_1$ and $y = z_2$:

$$I_{12} = \phi_1(b) \Phi_1\left(\frac{a+\rho b}{\sqrt{1-\rho^2}}\right) + \rho \phi_1(a) \Phi_1\left(-\frac{b+\rho a}{\sqrt{1-\rho^2}}\right)$$

This final formula for I_{12} is one of the key terms in calculating the expectation of the product of ReLUs.

- CALCULATION OF I_{22} (CROSS-MOMENT)

The calculation of I_{22} is the most complex part. The final result is :

$$I_{22} = \int_0^\rho I_{11} + \phi_2(a, b; \rho)$$

The proof of this identity was a central point of the discussion. The most rigorous method consists of showing the equality for $\rho = 0$, and then the equality of the derivatives with respect to ρ for both sides of the equation.

- FINAL FORMULA

By assembling all the terms, we obtain the final algebraic formula :

$$\mathbb{E}[\dots] = (\mu_1\mu_2 + \rho\sigma_1\sigma_2)I_{11} + \mu_1\sigma_2I_{12} + \mu_2\sigma_1I_{21} + \sigma_1\sigma_2\phi_2(a, b; \rho)$$

B.1 COMPUTING ReLU PRODUCT EXPECTATIONS

When analyzing the Neural Tangent Kernel (NTK) of a single-layer network, we need to compute expectations of the form :

$$\mathbb{E}[\text{ReLU}(X_1)\text{ReLU}(X_2)]$$

where (X_1, X_2) follows a bivariate normal distribution with parameters $(\mu_1, \sigma_1, \mu_2, \sigma_2, \rho)$.

For deterministic input vectors x_1 and x_2 , we define the following quantities :

- $\sigma_1 = \|x_1\|$ (where $\|\cdot\|$ denotes Euclidean norm)
- $\sigma_2 = \|x_2\|$
- $\rho = \cos(x_1, x_2) = \frac{\langle x_1, x_2 \rangle}{\|x_1\| \|x_2\|}$

The network outputs follow a Gaussian process characterized by the covariance matrix :

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

$$\begin{matrix} \sigma_1^2 \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 \sigma_2^2 \end{matrix}$$

B.2 DECOMPOSITION

We recall the expression of mu The approach is to standardize the variables $(Z_i = (X_i - \mu_i)/\sigma_i)$ and decompose the integral into four terms :

$$\mathbb{E}[\dots] = \sigma_1\sigma_2 I_{22} + \mu_1\sigma_2 I_{12} + \mu_2\sigma_1 I_{21} + \mu_1\mu_2 I_{11}$$

where the integrals I_{ij} are defined over the domain $z_1 > a$ and $z_2 > b$, with $a = -\mu_1/\sigma_1$ and $b = -\mu_2/\sigma_2$.

- $I_{11} = \int_a^\infty \int_b^\infty \phi_2(z_1, z_2; \rho) dz_2 dz_1$
- $I_{21} = \int_a^\infty \int_b^\infty z_1 \phi_2(z_1, z_2; \rho) dz_2 dz_1$
- $I_{12} = \int_a^\infty \int_b^\infty z_2 \phi_2(z_1, z_2; \rho) dz_2 dz_1$
- $I_{22} = \int_a^\infty \int_b^\infty z_1 z_2 \phi_2(z_1, z_2; \rho) dz_2 dz_1$

B.3 CALCULATION OF THE COMPONENTS

• CALCULATION OF I_{11} (PROBABILITY)

I_{11} is the probability $P(Z_1 > a, Z_2 > b)$. It is calculated via the cumulative distribution function (CDF) of the standard bivariate normal distribution, Φ_2 .

$$I_{11} = P(Z_1 > a, Z_2 > b) = P(Z_1 < -a, Z_2 < -b) = \Phi_2(-a, -b; \rho) = \Phi_2\left(\frac{\mu_1}{\sigma_1}, \frac{\mu_2}{\sigma_2}; \rho\right)$$

C

UNSUCCESSFUL ATTEMPTS TO CALCULATE I_{22}

Several approaches to calculate I_{22} proved to be dead ends.

C.1 VIA THE IDENTITY ON THE DERIVATIVE OF ϕ_2

Let's first prove the identity for the partial derivatives of the 2D normal PDF ϕ_2 . For a bivariate normal distribution with correlation ρ :

$$\frac{\partial \phi_2}{\partial z_1} = -\frac{z_1 - \rho z_2}{1 - \rho^2} \phi_2(z_1, z_2; \rho)$$

$$\frac{\partial \phi_2}{\partial z_2} = -\frac{z_2 - \rho z_1}{1 - \rho^2} \phi_2(z_1, z_2; \rho)$$

Using these identities, we can write :

$$z_1 \phi_2 = -\rho z_2 \phi_2 - (1 - \rho^2) \frac{\partial \phi_2}{\partial z_1}$$

Integrating from a to ∞ with respect to z_1 :

$$\int_a^\infty z_1 \phi_2 dz_1 = -\rho z_2 \int_a^\infty \phi_2 dz_1 + (1 - \rho^2) \phi_2(a, z_2; \rho)$$

Attempting to use this for I_{22} , we integrate $z_1 z_2 \phi_2$ over the region $[a, \infty) \times [b, \infty)$:

$$I_{22} = \int_b^\infty z_2 \left(\int_a^\infty z_1 \phi_2 dz_1 \right) dz_2$$

Substituting the identity :

$$I_{22} = -\rho \int_b^\infty z_2^2 \left(\int_a^\infty \phi_2 dz_1 \right) dz_2 + (1 - \rho^2) \int_b^\infty z_2 \phi_2(a, z_2; \rho) dz_2$$

This approach becomes intractable because : 1. The first term involves a higher-order moment z_2^2 2. The second term requires evaluating another complex integral 3. Each attempt to simplify leads to terms of equal or greater complexity

Therefore, while the partial derivative identities are mathematically elegant, they do not provide a practical path to computing I_{22} when beta is non 0.

D

SPECIAL CASE : ZERO MEANS ($\mu_1 = \mu_2 = 0$)

In this case, the "from scratch" derivation leads to an elegant formula.

$$\mathbb{E}[\neg(X_1) \neg(X_2)] = \sigma_1 \sigma_2 \int_0^\infty \int_0^\infty z_1 z_2 \phi_2(z_1, z_2; \rho) dz_1 dz_2$$

The problem reduces to calculating $I_{22}(0, 0; \rho) = \mathbb{E}[\neg(Z_1) \neg(Z_2)]$. The cleanest derivation uses Price's theorem :

1. $\frac{\partial \mathbb{E}[g]}{\partial \rho} = \mathbb{E}\left[\frac{\partial^2 g}{\partial z_1 \partial z_2}\right] = \mathbb{E}[\mathcal{K}(z_1 > 0, z_2 > 0)] = P(Z_1 > 0, Z_2 > 0) = \frac{1}{2\pi} \arccos(-\rho)$.
2. The integration constant is $\mathbb{E}[g]_{\rho=0} = \mathbb{E}[\neg(Z_1)]\mathbb{E}[\neg(Z_2)] = \left(\frac{1}{\sqrt{2\pi}}\right)^2 = \frac{1}{2\pi}$.
3. By integrating $\frac{\partial \mathbb{E}[g]}{\partial \rho}$ from 0 to ρ :

$$\begin{aligned} \mathbb{E}[g](\rho) - \mathbb{E}[g](0) &= \int_0^\rho \frac{1}{2\pi} \arccos(-t) dt \\ &= \frac{1}{2\pi} \left[t \arccos(-t) + \sqrt{1-t^2} \right]_0^\rho \\ &= \frac{1}{2\pi} (\rho \arccos(-\rho) + \sqrt{1-\rho^2} - 1) \end{aligned}$$

4. By adding the constant $\mathbb{E}[g](0) = 1/(2\pi)$, we get :

$$\mathbb{E}[\neg(Z_1) \neg(Z_2)] = \frac{1}{2\pi} (\sqrt{1-\rho^2} + \rho \arccos(-\rho))$$

The final formula for variances σ_1^2, σ_2^2 is :

$$\mathbb{E}[\neg(X_1) \neg(X_2)] = \frac{\sigma_1 \sigma_2}{2\pi} (\sqrt{1-\rho^2} + \rho \arccos(-\rho))$$

D.1 EXPLICIT FORMULAS FOR Σ AND $\dot{\Sigma}$

For layer ℓ , the general formulas are :

$$\begin{aligned} \Sigma^{(\ell)}(x, x') &= \mathbb{E}_{w,b} \left[\sigma(w^\top h^{(\ell-1)}(x) + \beta b) \sigma(w^\top h^{(\ell-1)}(x') + \beta b) \right] \\ \dot{\Sigma}^{(\ell)}(x, x') &= \mathbb{E}_{w,b} \left[\dot{\sigma}(w^\top h^{(\ell-1)}(x) + \beta b) \dot{\sigma}(w^\top h^{(\ell-1)}(x') + \beta b) w^\top w \right] \end{aligned}$$

When $\beta = 0$, these simplify to :

$$\begin{aligned} \Sigma^{(\ell)}(x, x') &= \mathbb{E}_w \left[\sigma(w^\top h^{(\ell-1)}(x)) \sigma(w^\top h^{(\ell-1)}(x')) \right] \\ \dot{\Sigma}^{(\ell)}(x, x') &= \mathbb{E}_w \left[\dot{\sigma}(w^\top h^{(\ell-1)}(x)) \dot{\sigma}(w^\top h^{(\ell-1)}(x')) w^\top w \right] \end{aligned}$$

Note that for $\ell > 1$, both kernels become random fields due to the randomness in $h^{(\ell-1)}$.

D.2 DIFFERENTIAL AND INTEGRAL FORM

The theorem relates the derivative of the expectation of a function g with respect to the covariance σ_{12} to the expectation of the second derivative of g . For $g = \mathcal{J}(X_1) \mathcal{J}(X_2)$, we have $\frac{\partial^2 g}{\partial x_1 \partial x_2} = \mathbb{1}(X_1 > 0, X_2 > 0)$.

$$\frac{\partial \mathbb{E}[\mathcal{J}(X_1) \mathcal{J}(X_2)]}{\partial \sigma_{12}} = \mathbb{E}[\mathbb{1}(X_1 > 0, X_2 > 0)] = I_{11}$$

This leads to the integral formulation :

$$\mathbb{E}[\mathcal{J}(X_1) \mathcal{J}(X_2)] = \int I_{11}(\sigma_{12}) d\sigma_{12} + \text{Constant}$$

D.3 CALCULATING THE CONSTANT (INDEPENDENT CASE $\rho = 0$)

The constant is the value of the expectation when $\rho = 0$, i.e., when X_1 and X_2 are independent.

$$C = \mathbb{E}[\mathcal{J}(X_1)] \cdot \mathbb{E}[\mathcal{J}(X_2)]$$

The expectation of a single ReLU is :

$$\mathbb{E}[\mathcal{J}(X)] = \sigma \phi_1\left(\frac{\mu}{\sigma}\right) + \mu \Phi_1\left(\frac{\mu}{\sigma}\right)$$

The constant is therefore the product of two such terms :

$$C = \left[\sigma_1 \phi_1\left(\frac{\mu_1}{\sigma_1}\right) + \mu_1 \Phi_1\left(\frac{\mu_1}{\sigma_1}\right) \right] \times \left[\sigma_2 \phi_1\left(\frac{\mu_2}{\sigma_2}\right) + \mu_2 \Phi_1\left(\frac{\mu_2}{\sigma_2}\right) \right]$$

This is the implementation of the constant in code, this is fairly long for non zeros beta, that's why we focus on the hypothesis with beta = 0. This in order to make experiments and proofs that MMNNs are superior to FCNNs for 2 hidden layer networks.

D.4 EXACT MOMENTS AND CORRELATION FOR KIBBLE VARIABLES

Let $(X_i, Y_i)_{i=1}^r$ be i.i.d. Gaussian pairs with zero mean, unit variances and correlation ρ , and define

$$U = \sum_{i=1}^r X_i^2, \quad V = \sum_{i=1}^r Y_i^2.$$

Then (U, V) follows Kibble's bivariate chi-square law with r degrees of freedom and correlation ρ . By Isserlis' (Wick's) theorem (or by expanding $\mathbb{E}[(X_i - \rho Y_i)^n (Y_i)^m]$ recursively) :

$$\mathbb{E}[X_i^2] = 1, \quad \text{Var}(X_i^2) = 2, \quad \mathbb{E}[X_i^2 Y_i^2] = 1 + 2\rho^2, \quad (X_i^2, Y_i^2) = 2\rho^2.$$

Using independence across i :

$$\mathbb{E}[U] = r, \quad \text{Var}(U) = 2r, \quad \mathbb{E}[V] = r, \quad \text{Var}(V) = 2r,$$

$$(U, V) = \sum_{i=1}^r (X_i^2, Y_i^2) = 2r \rho^2,$$

$$(U, V) = \frac{(U, V)}{\sqrt{\text{Var}(U) \text{Var}(V)}} = \frac{2r \rho^2}{\sqrt{(2r)(2r)}} = \rho^2,$$

$$\mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V] + (U, V) = r^2 + 2r \rho^2.$$

D.5 PRODUCT OF SQUARE ROOTS OF KIBBLE VARIABLES

Let $S = \sqrt{U}\sqrt{V} = \|X\|\|Y\|$. Due to correlation between U and V , we cannot directly multiply expectations. Instead, using Cauchy-Schwarz inequality :

$$\mathbb{E}[S^2] = \mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V] + (U, V) = r^2 + 2r \rho^2$$

Therefore :

$$\mathbb{E}[S] \leq \sqrt{\mathbb{E}[S^2]} = r \sqrt{1 + \frac{2\rho^2}{r}}$$

The variance can be estimated similarly by expanding $(S - \mathbb{E}[S])^2$ and using correlation terms :

$$\text{Var}(S) \approx \frac{r}{2}(1 + \rho^2) + O(1)$$

This suggests S/r approaches a distribution with mean $O(\sqrt{1 + \frac{2\rho^2}{r}})$ and variance of order $O(1/r)$.

D.6 INDEPENDENCE PROPERTIES AND ASYMPTOTIC BEHAVIOR

The orientation variable r is independent of magnitudes (U, V) . For the normalized product S/r , we have :

$$\frac{S}{r} = \frac{\|X\|\|Y\|}{r} \xrightarrow{r \rightarrow \infty} 1$$

This convergence occurs with fluctuations of order $O(1/\sqrt{r})$, showing that in high dimensions, the normalized product concentrates around 1 regardless of the correlation ρ .

E

EXPÉRIENCES NUMÉRIQUES

(paragraphe)